

How to solve the ciphers

Cipher	What are you given?	How to encrypt	How to decrypt	Remove duplicates?	Main thing to remember
Caesar Cipher	A shift number n	Move every plaintext letter forward by n	Move every ciphertext letter backward by n	No	The shift can be any given number
ROT13	Nothing extra; shift is fixed	Move every letter 13 positions	Move every letter 13 positions again	No	It's basically Caesar with $n = 13$
Keyword Cipher	A keyword/phrase	Build the cipher alphabet using the keyword, then map top $\rightarrow$ bottom	Use the same table but map bottom $\rightarrow$ top	Keyword only	Remove duplicates from the keyword , NOT the message
Playfair Cipher	Keyword + 5×5 matrix	Split plaintext into pairs , then apply row/column/rectangle rules	Reverse the same rules	From the keyword when building the matrix	Works with 2 letters at a time
Vigenère Cipher	Keyword + Vigenère table	Repeat keyword to plaintext length; row = plaintext, column = key , intersection = ciphertext	Row = key , find ciphertext in row, corresponding column = plaintext	No	Repeat the keyword until it matches message length
One-Time Pad	Random shift keys $k_1, k_2, \dots, k_n$	Apply a different random shift to each plaintext letter	Reverse each corresponding shift	No	Each shift key is random
Hill Cipher	An invertible matrix	Convert letters to numbers, group into blocks, multiply by key matrix mod 26	Multiply blocks by inverse key matrix mod 26	No	This one is matrix math